

CUQ

1. If 'n' is a positive integer then the number of terms in the expansion of $(x+a)^n$ is

1) n
2) n + 1
3) n + 2
4) Infinitely
2. In any Binomial expansion, with positive integral index, number of terms is

1) ≥ 1
2) ≥ 2
3) ≥ 3
4) ≥ 4
3. ${}^nC_r =$

1) $\frac{n!}{(n-r)!}$
2) $\frac{n!}{(n-r)!r!}$
3) $\frac{(n-r)!}{n}$
4) $\frac{(n-r)!}{(n-r)!}$
4. ${}^nC_0 = {}^nC_n =$

1) 0
2) - 1
3) n
4) 1
5. ${}^nP_r =$

1) $\frac{n!}{(n-r)!}$
2) $\frac{n!}{(n-r)!r!}$
3) $\frac{(n-r)!}{n}$
4) $\frac{(n-1)!}{(n-r)!}$
6. The general term in the expansion of $(x+a)^4$ is

1) ${}^4C_r x^{4-r}$
2) ${}^nC_r x^r a^{n-r}$
3) ${}^4C_r x^{4-r} a^r$
4) $x^{4-r} a^r$
7. Which term is the middle term(s) in the expansion of $(a+b)^n$, if n is even is

1) $\left(\frac{n}{2}+1\right)^{th}$ term
2) $\left(\frac{n+1}{2}\right)^{th}$ term
3) $\left(\frac{n}{2}\right)^{th}$ term
4) $\left(\frac{n}{2}-2\right)^{th}$ term
8. The middle term(s) in the expansion of $(x+2)^2$ is

1) 2x
2) 4
3) x^2
4) 4x
9. The middle term(s) in the expansion of $(x+1)^3$ is

1) 3x
2) $3x^2$
3) x^3
4) Both (1) & (2)
10. The term independent of x in $\left(x+\frac{1}{x}\right)^2$ is

1) 1
2) 2
3) $\frac{1}{2}$
4) 2x

JEE MAIN

LEVEL - 1

Single Correct Choice Type:

- If $n! = 120$, then the value of n is
 1) 3 2) 4 3) 6 4) 5
- If ${}^nC_{n-3} = 10$, then $n =$
 1) 4 2) 5 3) 6 4) 7
- If ${}^{n+1}C_3 = 4 \cdot {}^nC_2$, then $n =$
 1) 12 2) 10 3) 16 4) 11
- If ${}^nC_{12} = {}^nC_8$, then $n =$
 1) 8 2) 12 3) 20 4) 4
- Expand $(x+6)^5$.
 1) $x^5 + 30x^4 + 360x^3 - 2160x^2 - 6480x + 7776$
 2) $x^5 - 30x^4 + 360x^3 - 2160x^2 - 6480x + 7776$
 3) $x^5 + 30x^4 + 360x^3 + 2160x^2 + 6480x + 7776$
 4) $x^5 + 60x^4 + 360x^3 + 260x^2 + 6480x + 7776$
- The 6th term of $(2+x)^{10}$ is
 1) ${}^{10}C_5 5^2 x^5$ 2) ${}^{10}C_5 2^5 x^5$ 3) ${}^{10}C_2 2^2 x^5$ 4) ${}^{10}C_2 5^2 x^5$
- Middle term in the expansion $\left(x - \frac{1}{2x}\right)^{10}$ is
 1) $\frac{105}{32x^2}$ 2) $\frac{63}{8}$ 3) $\frac{-105}{32x^2}$ 4) $\frac{-63}{8}$
- The term independent of x in $\left(x^2 - \frac{1}{x}\right)^9$ is
 1) 84 2) 96 3) 112 4) 125

JEE MAIN

LEVEL - 2

- If ${}^{15}C_{r+3} = {}^{15}C_{2r-3}$, then $r =$
 1) 4 or 7 2) 5 or 6 3) 7 or 9 4) 9 or 10

10. If ${}^nP_r = 1320$, ${}^nC_r = 220$, then the value of 'r' is

- 1) 3 2) 4 3) 5 4) 6

11. The 12th term of $(2x-1)^{13}$ is

- 1) $-{}^{13}C_2(2x)^2$ 2) ${}^{13}C_2(2x)^2$ 3) $-{}^{12}C_2(2x)^2$ 4) ${}^{12}C_2(2x)^2$

12. The coefficient of x^{-4} in $\left(2x - \frac{3}{x}\right)^8$ is

- 1) ${}^8C_3 2^6 3^6$ 2) ${}^8C_4 2^4 3^4$ 3) ${}^8C_6 2^3 3^5$ 4) ${}^8C_6 2^2 3^6$

JEE MAIN

LEVEL - 3

13. If ${}^{28}C_r : {}^{24}C_{r-4} = 225 : 11$, then r =

- 1) 13 2) 14 3) 15 4) 16

14. Number of non zero terms in the expansion of $(x+a)^{100} + (x-a)^{100}$ is

- 1) 41 2) 51 3) 61 4) 71

15. The value of ${}^{10}C_5 + 2 \cdot {}^{10}C_4 + {}^{10}C_3$ is

- 1) ${}^{11}C_4$ 2) ${}^{11}C_5$ 3) ${}^{12}C_5$ 4) ${}^{12}C_4$

JEE MAIN

LEVEL - 4

16. If ${}^{(n+1)}P_5 : {}^nP_6 = 2 : 7$, then the value of n is

- 1) 13 2) 11 3) 15 4) 17

17. The ratio of T_{12} and T_{13} in the expansion of $\left(x + \frac{1}{y}\right)^{23}$ is

- 1) $xy : 4$ 2) $\frac{xy}{2}$ 3) $xy : 1$ 4) $xy : 3$

18. The value of ${}^{50}C_4 + \sum_{r=1}^6 {}^{(56-r)}C_3$ is

- 1) ${}^{55}C_4$ 2) ${}^{55}C_3$ 3) ${}^{56}C_3$ 4) ${}^{56}C_4$

JEE MAIN

LEVEL - 5

19. If the 5th term of $\left(\frac{q}{2x} - px\right)^8$ is 1120 and $p+q=5$, $p>q>0$, then $p=$
- 1) 3 2) 6 3) 4 4) 7
20. If T_{r+1} is the independent term of x in $\left(ax + \frac{6}{x}\right)^{14}$, then r is
- 1) 5 2) 7 3) 14 4) 6
21. 3rd term from the end of the expansion in $\left(\frac{3x}{5} - \frac{5}{2x}\right)^8$ is
- 1) $\frac{39375}{16.x^4}$ 2) $\frac{37395}{16.x^4}$ 3) $\frac{39537}{16.x^4}$ 4) $\frac{39375}{32.x^4}$

JEE ADVANCED

LEVEL - 1

Multi Correct Choice Type:

22. If $n = 5 \times 4 \times 3 \times 2 \times 1$, then $n =$
- 1) 5! 2) 120 3) $5 \times 4!$ 4) $20 \times 3!$

Comprehension Type:

If ${}^nC_r = {}^nC_s$, then $r = s$ and $n = r + s$

23. If ${}^nC_4 = {}^nC_8$, then ${}^nC_2 =$
- 1) 64 2) 65 3) 66 4) 62
24. If ${}^nC_{24} = {}^nC_{35}$, then ${}^{60}C_n =$
- 1) 59 2) 60 3) 75 4) -60
25. If ${}^{30}C_{2r+3} = {}^{30}C_{3r-3}$, then $r =$
- 1) 6 2) 30 3) 2 4) 5

Numerical Value Type:

26. If ${}^nC_4 = {}^nC_0$, then $n =$ _____

Matrix Match Type:27. **Column-I****Column-II**a) The middle term in $\left(\frac{3x}{7} - 2y\right)^{10}$ is

p) ${}^8C_4(14)^4(pq)^4$

b) The middle term in $(7p + 2q)^8$ is

q) ${}^{20}C_{10}(15)^{10} \cdot a^{10}$

c) The middle term in $(4x^2 + 5x^3)^{17}$ is

r) ${}^{-10}C_5\left(\frac{6}{7}\right)^5 x^5 \cdot y^5$

d) The middle term in $\left(\frac{3}{a^3} + 5a^4\right)^{20}$ is

s) ${}^{17}C_9 \cdot 4^8 5^9 \cdot x^{43}$

Statement Type:28. **Statement-I:** In the expansion $(x+a)^n$, r^{th} term from the end $= (n-r+2)^{th}$ term from the beginning.**Statement-II:** In the expansion $(3+y)^{10}$, 7^{th} term from the end is 5^{th} term from the beginning.

- 1) Statement-I true and Statement-II is true
- 2) Statement-I true and Statement-II is false
- 3) Statement-I false and Statement-II is true
- 4) Statement-I false and Statement-II is false

JEE ADVANCED**LEVEL-2&3****Multi Correct Choice Type:**29. Coefficient of x^2 in the binomial expansion $(1-x)^2$ is

1) $\frac{2^n}{2!}$

2) 1

3) $\frac{n^2 + 3n + 2}{(n+2)(n+1)}$

4) $\frac{n^2 - 1}{(n-1)(n+1)}$

Matrix Match Type:30. **Column - I****Column - II**

a) ${}^nC_r =$

p) $\frac{n}{r} \cdot {}^{n-1}C_{r-1}$

b) ${}^nP_r =$

q) ${}^{n-1}C_r + {}^{n-1}C_{r-1}$

c) If T_{r+1} is middle term of $(x+a)^n$, n is odd, then r is

r) $\frac{n+1}{2}$

d) $\frac{{}^nP_r}{r!} =$

s) $\frac{n-1}{2}$

t) ${}^{n-1}P_r + r \cdot {}^{n-1}P_{r-1}$

Numerical Value Type:

31. The number of terms in the expansion of $(1 + 5\sqrt{2}x)^9 + (1 - 5\sqrt{2}x)^9$ is _____

JEE ADVANCED**LEVEL-4&5****Comprehension Type:**

If a and b are real numbers, then for all $n \in \mathbb{N}$

$$(a + b)^n = {}^nC_0 a^n b^0 + {}^nC_1 a^{n-1} b^1 + {}^nC_2 a^{n-2} b^2 + \dots + {}^nC_r a^{n-r} b^r + \dots + {}^nC_{n-1} a^1 b^{n-1} + {}^nC_n a^0 b^n$$

General term in $(x + a)^n$ is $T_{r+1} = {}^nC_r x^{n-r} \cdot a^r$ ($0 \leq r \leq n$)

The co-efficient of ' x^k ' in the expansion of $\left(ax^p + \frac{b}{x^q}\right)^n$ is ${}^nC_r a^{n-r} b^r$, where $r = \frac{np - k}{p + q}$

32. If the co-efficient of ' x ' in $\left(x^2 + \frac{k}{x}\right)^5$ is 270, then $k =$

- 1) 0 2) 1 3) 2 4) 3

33. The term independent of x in $\left(x^3 - \frac{1}{x}\right)^8$ is

- 1) 28 2) 38 3) 56 4) 18

34. If the coefficients of x^2 and x^3 in $(3 + kx)^9$ are equal, then $k =$

- 1) $\frac{9}{7}$ 2) $\frac{7}{9}$ 3) $\frac{3}{4}$ 4) $\frac{4}{3}$

Numerical Value Type:

35. The sum of the coefficients of x^{32} and x^{-17} in $\left(x^4 - \frac{1}{x^3}\right)^{15}$ is _____

SEEKING TO DESCRIBE

36. If ${}^nP_3 = 1320$, then find the value of n .

37. Find the value of ${}^{47}C_4 + \sum_{j=1}^5 ({}^{52-j}C_3)$?

38. If ${}^nP_r = 1320$, ${}^nC_r = 220$, then find the values of n and r .

STUDENT ADDITIONAL TASK

- The second term in the expansion of $(7x-4y)^3$ is
 1) $588x^2y$ 2) $688x^2y$ 3) $-588x^2y$ 4) $-688x^2y$
- If $n = 10$, then the value of ${}^{n+1}C_3 - {}^{n-1}C_3 =$
 1) n^2 2) $(n-1)^2$ 3) $(n+1)^2$ 4) $2n^2$
- If ${}^nC_4 = {}^nC_s$, then
 1) $s = 4$ 2) $n = s + 4$ 3) $n = s$ 4) $n = 4$
- Expand $\left(\frac{2x}{y} + \frac{5y}{x}\right)^6$
- If ${}^nC_r = {}^nC_{r-1}$ and ${}^nP_{r+1} = 10 {}^nP_r$, then $(n, r) =$
 1) $(19, 9)$ 2) $(20, 10)$ 3) $(20, 9)$ 4) $(19, 10)$
- If the 3rd term of $\left(\frac{1}{x} + x^{\log_{10} x}\right)^5$ is 1, then $x =$
 1) 1 2) 10 3) 100 4) $\frac{1}{\sqrt{10}}$
- If $C(2n, 3) : C(n, 2) = 12 : 1$, then $n =$
 1) 4 2) 5 3) 6 4) 8
- If ${}^{2n}C_3 : {}^nC_2 = 20 : 3$, then $n =$
 1) 5 2) 4 3) 6 4) 3
- Middle term in the expansion of $\left(2x - \frac{3}{x}\right)^{15}$ is
 1) $-{}^{15}C_7 2^8 3^7 x$ 2) ${}^{15}C_7 2^8 3^7 x$ 3) $-{}^{15}C_6 2^8 3^7 x$ 4) ${}^{15}C_6 2^8 3^7 x$
- Middle terms in the expansion of $(x+a)^9$ are
 1) t_4, t_5 2) t_5, t_6 3) t_6, t_7 4) t_7, t_8

11. The middle term in the expansion of $\left(x + \frac{1}{2x}\right)^{2n}$ is
- 1) $\frac{1.3.5....(2n-3)}{n!}$ 2) $\frac{1.3.5....(2n+1)}{n!}$ 3) $\frac{1.3.5....(2n-1)}{n!}$ 4) $\frac{1.3.5....(2n+3)}{n!}$
12. Number of terms in the expansion of $(1+2x+x^2)^{20}$ is
- 1) 20 2) 21 3) 40 4) 41
13. The value of $\frac{9!-6!}{4!}$ is
- 1) 15000, 2) 15090 3) 15060 4) 15030
14. The term independent of x in the expansion of $\left(\frac{x+1}{x^{2/3}-x^{1/3}+1} - \frac{x-1}{x-x^{1/2}}\right)^{10}$ is
- 1) $T_5 = 210$ 2) $T_5 = -210$ 3) $T_5 = 180$ 4) $T_5 = -180$
15. If the coefficients of $5^{th}, 6^{th}, 7^{th}$ terms in the expansion of $(1+x)^n$ are in A.P, then n=
- 1) 7 2) 14 3) 7 (or) 14 4) 8
16. The ratio of 3^{rd} and 4^{th} terms of $\left(x + \frac{2}{3}x^2\right)^7$ is
- 1) x^3 2) $3x^3$ 3) $\frac{0.9}{x}$ 4) $0.3x^3$
17. The coefficient of x^p in the expansion of $\left(x^2 + \frac{1}{x}\right)^{2n}$ is _____ (when exists).
- 1) ${}^{2n}C_{\frac{4n+p}{3}}$ 2) ${}^{2n}C_{\frac{4n-p}{3}}$ 3) ${}^{2n}C_{\frac{n+p}{3}}$ 4) ${}^nC_{\frac{n+p}{3}}$
18. ${}^{2n}C_{n+1} + 2 \cdot {}^{2n}C_n + {}^{2n}C_{n-1} =$
- 1) ${}^{(2n+2)}C_{n+1}$ 2) ${}^{(n+1)}C_{n+1}$ 3) nC_n 4) ${}^{(2n+1)}C_{n+1}$
19. $\frac{2^n (1.3.5.....(2n-1))}{n!} =$
- 1) ${}^{2n}P_n$ 2) ${}^{2n}C_n$ 3) ${}^{3n}C_n$ 4) ${}^{4n}C_{2n}$
20. The value of ${}^{95}C_4 + \sum_{j=1}^5 ({}^{100-j}C_3) =$
- 1) ${}^{99}C_5$ 2) ${}^{100}C_4$ 3) ${}^{99}C_4$ 4) ${}^{100}C_5$
